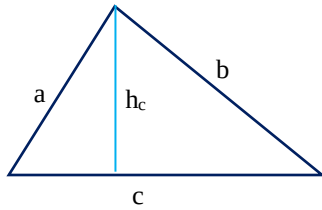




Parts of General Triangle

1. Height of a general triangle

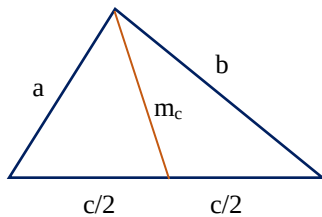


$$h_c = \frac{2\sqrt{s(s-a)(s-b)(s-c)}}{c}$$

where: $s = \frac{a+b+c}{2}$

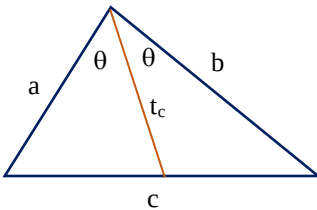
Note: h_c is perpendicular to side c.

2. Median of a general triangle



$$m_c = \frac{1}{2}\sqrt{2a^2 + 2b^2 - c^2}$$

3. Angle bisector of a general triangle



$$t_c = \sqrt{ab \left[1 - \left(\frac{c}{a+b} \right)^2 \right]}$$

Important Properties of a Triangle

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1. The sum of two sides of a triangle is greater than the third side and their difference is less than the third side.
2. The perpendicular bisectors of the sides and the bisectors of the angles of a triangle, meet in points which are the center of the circumscribing circle the inscribed circle, respectively.

4. The altitudes of a triangle meet in a point.
5. The medians of a triangle are concurrent at a point which is $\frac{2}{3}$ of the distance from any vertex to the midpoint of the opposite sides.

Radius of Inscribed Circle and Radius of Circumscribing Circle

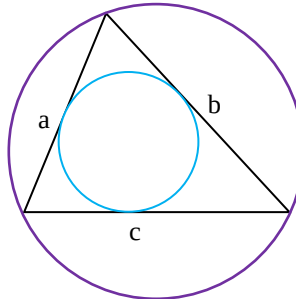
Let:

a, b and c = sides of triangle

r = radius of inscribed circle

R = radius of circumscribing circle

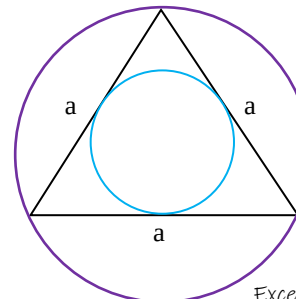
3. General triangle



$$r = \frac{A_{\text{triangle}}}{a+b+c}$$

$$R = \frac{abc}{4A_{\text{triangle}}}$$

2. Equilateral triangle

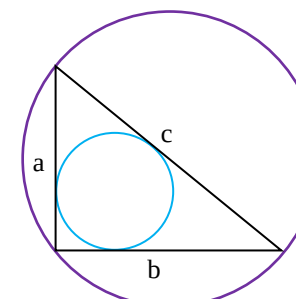


$$r = \frac{a\sqrt{3}}{6}$$

$$R = \frac{a\sqrt{3}}{3}$$

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1. Right triangle



$$r = \frac{ab}{a+b+c}$$

$$R = \frac{c}{2}$$

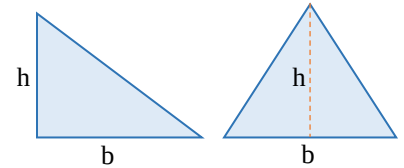


Did You Know?

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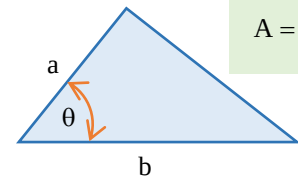
Area of Triangle

1. Given base and altitude



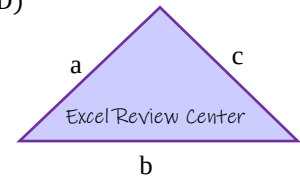
$$A = \frac{1}{2}bh$$

2. Given 2 sides and included angle



$$A = \frac{1}{2}ab \sin \theta$$

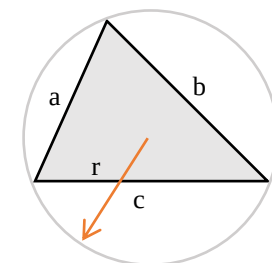
3. Given 3 sides
Use Heron's formula, named after Heron of Alexandria (1st Century AD)



$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

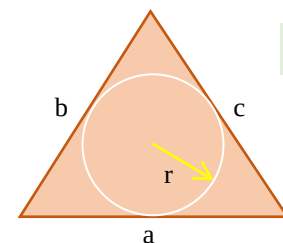
where: $s = \frac{a+b+c}{2}$

4. Triangle inscribed in a circle



$$A = \frac{abc}{4r}$$

5. Triangle circumscribing a circle



$$A = rs$$

where: $s = \frac{a+b+c}{2}$